

Pratham Pal

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EDUCATION

Indian Institute of Information Technology, Allahabad

Prayagraj, Uttar Pradesh

Bachelor of Technology in Electronics and Communication Engineering

Nov 2022 – Present

- CGPA: **8.0/10.0**

- Relevant Coursework: Data Structures & Algorithms, Computer Networks, Operating Systems, DBMS, OOP, System Design

WORK EXPERIENCE

Data Engineering Intern

Jan 2026 – Present

DealShare

- Developed end-to-end data pipelines on **Databricks** to monitor real-time user activity and large-scale data usage patterns
- Designed **ETL workflows** to ingest, transform, and load large volumes of data, enabling downstream analytics on user behavior and consumption metrics
- Refactored legacy **Mage**-based workflows to optimized **PySpark** jobs, improving throughput and reducing pipeline execution time significantly
- Optimized PySpark data processing jobs for efficient large-scale data handling, enhancing pipeline scalability and infrastructure reliability
- Applied **Python** and **SQL** for complex data transformation, cleansing, and exploratory analysis across multiple heterogeneous data sources

PROJECTS

Network Security Threat Detection | *Python, Scikit-learn, MLflow, DagsHub*

June 2025 | **GitHub**

- Architected a **modular ML pipeline** covering data ingestion, preprocessing, feature engineering, model training, and evaluation for improved maintainability and reusability
- Trained and compared **Random Forest**, **Gradient Boosting**, **AdaBoost** and **Logistic Regression** with systematic hyperparameter tuning across all models
- Achieved **F1-score: 0.974**, **Precision: 0.962**, **Recall: 0.986**, ensuring high reliability for real-world network threat detection
- Integrated **MLflow** and **DagsHub** for complete experiment tracking, model versioning, and reproducibility across all training runs

Image Segmentation Using U-Net and Mask R-CNN | *TensorFlow, Keras, Google Colab*

June 2024 | **GitHub**

- Implemented **U-Net** and **Mask R-CNN** deep learning architectures for pixel-level object detection and semantic segmentation
- Designed end-to-end pipeline for **data preprocessing, augmentation, training, and evaluation** on the Oxford-IIIT Pet Dataset
- Achieved **88% validation accuracy** with U-Net; obtained **IOU of 0.774** and **DICE score of 0.872** on the pet class

TECHNICAL SKILLS

- Programming Languages:** C++, C, Python, SQL
- Data Engineering:** Apache Spark (PySpark), Databricks, ETL Pipeline Development, Data Modeling, Mage
- ML & AI:** Scikit-learn, TensorFlow, Keras, PyTorch, MLflow, DagsHub, Hyperparameter Tuning
- Databases & Tools:** MySQL, Git, Jupyter Notebook, Google Colab
- Core Concepts:** Data Structures & Algorithms, OOP, System Design, Computer Networks, DBMS, Operating Systems

ACHIEVEMENTS & CERTIFICATIONS

- Selected in **Amazon ML Summer School 2024** among **85,000+** applicants, covering supervised learning, deep learning, and NLP (**Link**)
- Solved **1500+** problems across competitive programming platforms (Codeforces, CodeChef, LeetCode)
- Global Rank **222** in Codeforces Div3 Contest 1013 (**Link**) | Global Rank **229** in CodeChef Div2 Starters 193
- Global Rank **681** in LeetCode Biweekly Contest 161
- Completed **TensorFlow Developer Professional Certificate** by DeepLearning.AI on Coursera (**Link**)

CODING PROFILES

- Codeforces:** kalki_G (Max: 1673 – **Expert**) | **CodeChef:** kalki.g (Max: 1912 – **4 Star**) | **LeetCode:** kalki_G (Max: 1913 – **Knight**)

POSITION OF RESPONSIBILITY

AI/ML Wing Coordinator

2024 – 2025

Google Developer Student Club, IIT Allahabad

- Led the AI/ML wing by organizing workshops, hackathons, and study sessions to upskill **100+ students** in machine learning and data science
- Managed end-to-end event planning and cross-team coordination, consistently delivering activities on schedule and within scope
- Improved member participation and engagement through structured programs, communication strategies, and peer mentorship